Supporting Information

The price of absolute scale: degeneracy groups, surviving statistics, and the reference economics of cellular signalling

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This Supporting Information documents the complete computational output of the accompanying repository (`scale-free-framework`, release v1.0.1): the full analysis-script roster, every archived verdict table with its frozen decision lines, and the provenance of all third-party data. Numbers are transcribed verbatim from `results/`. Scripts are plain Python 3 (numpy/scipy/matplotlib/pandas); most are self-contained against `results/` and `data_external/`, except the per-embryo Bicoid anchor audit which requires the Nikolić et al. repository data (download instructions in `data_external/PROVENANCE.md`).

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1 Pipeline map: scripts and outputs

Script	Computes	Archived output
PdPC instance		
code04_pdpc_identifiability_audit	identifiability audit of the PdPC circuit	figures/tables
code10_asymmetric_Km_degeneracy_group	asymmetric-K_m degeneracy group	figures/tables
code11_discrimination_map_v09	discrimination map	figures
Census systems (NF-κB, Msn2)		
code06_nfkB_negative_feedback_ oscillator_audit	NF-κB oscillator audit	figures/tables
code08_msn2_excitable_frequency_ encoding	Msn2 excitable frequency encoding	figures/tables
code09_nfkB_spike_mode_audit	NF-κB spike-mode audit	figures/tables
code12_nfkB_GLM_prediction_test_ switchlike_IkB	GLM prediction test, switch-like IκB	figures/tables
code15_nfkB_saturation_transient_ test	saturation transient test	figures/tables
Blind adjudications		
code44/44b	P2 amplitude-stratification adjudication (Msn2) + CI recompute	P2_verdict_summary.csv, P2_msn2_stratification_units.csv
code45	P3 free-statistic sufficiency (NF-κB spatial gradient)	P3_verdict_summary.csv
code46	P4 ERK re-adjudication (GPCR-KTR, trajectory level)	P4_verdict_summary.csv
code48	P6 FCD attribution (NF-κB sequential grid)	P6_verdict_summary.csv
Morphogen instance		
p9_bcd_anchor_audit.py	P9: absolute-anchor test of the Bcd profile (581 embryos)	p9_bcd_anchor_results.json, p9_2_upgrade_results.json
audit_bergmann2010_axis12.py	Bergmann 2010 axes-1/2 audit	records/logs/audit_bergmann2010_axis12.txt
Yeast positive control		
score_markers_v2_authordata.py	Hog1 osmotic set-point markers from author data	records/logs/score_markers_v2_authordata.txt

Script	Computes	Archived output
Cascade instance		
cascade_audit_gpcr_ca.py	GPCR→Ca cascade audit; per-experiment Hill fits	cascade_audit_results.json, cascade_hill_fits_methods_ladder.json
p10_gain_confinement_audit.py	P10: gain-distribution confinement across systems	p10_gain_confinement_results.json, p10_supplementary_arms_results.json
Tumour chain P11–P16		
p11_counting_law_cross_cellline_ audit.py	P11 counting-law cross-cell-line audit	digitised arms CSVs
p12a_erk_violin_digitize.py	ERK violin digitisation (H1299)	P12a_erk_baseline_cv.csv
p12b_*.py (5 scripts)	P12b resistance zero-direction: EGFR DMS, ABL catalytic axis, extensions	p12b_*.json, p12b_egfr_dms_variants.csv, P12b_*.csv
p13_mutual_info_audit.py, p13_robustness_recheck.py	P13 channel capacity (mutual information)	p13_mi_results.json, p13_robustness.json, p13_fig6_decay.json
p14_shape_decoding_audit.py	P14 downstream mis-reading / shape decoding	p14_results.json, p14_pergene_log2R.csv
p15_tcga_decoupling_audit.py	P15 TCGA tissue-level decoupling	p15_tcga_results.json, four per-sample CSVs
Manuscript figures (all in manuscript/)		
make_fig1_schematic.py	Fig. 1 framework schematic (derived from the umbrella edition draw_ed_figures.py, EDFig1 section)	fig1_schematic.png
make_fig2_pdpc.py	Fig. 2 PdPC static sensor (renamed copy of code/make_fig2.py)	fig2_pdpc.png
make_fig3.py	Fig. 3 PdPC full audit panels	fig3_pdpc_full.png
make_fig4.py	Fig. 4 instance atlas	fig4_instances.png
make_fig5_ledger.py	Fig. 5 confirmation/falsification ledger (renamed copy of code/make_fig1.py)	fig5_ledger.png

Run logs for every analysis script in this table are archived under records/logs/ in the repository; scripts whose output is console-only have their stdout captured there as .txt, and the figure scripts in manuscript/ now read only data_external/ and results/ of the repository.

2 Blind adjudication campaign: verdicts against frozen lines

2.1 P2 — amplitude stratification (Msn2, 7 promoters)

Clause	Frozen line	Verdict
P2-1	$\Delta_{\text{mol}} \geq +0.03$ in $\geq 6/7$ promoters	**Falsified** (0/7)
P2-2	$\Delta_{\text{AU}} \leq 0.01$ and $< \Delta_{\text{mol}}$ in $\geq 6/7$	**Falsified** (0/7)
P2-3	$\lvert\Delta_{\text{time}}\rvert \leq 0.01$ (k=3) in $\geq 6/7$	**Hit** (6/7; 7/7 at k=2; 6/7 at k=4, DCS2 $= -0.0102$)

Per-promoter unit statistics (183 KB) in `P2_msn2_stratification_units.csv`; per-promoter clause values in `P2_verdict_summary.csv` (e.g. ALD3: $\Delta_{\text{mol}} = -0.0006$, $\Delta_{\text{AU}} = +0.0013$, $\Delta_{\text{time}}(k{=}3) = -0.0046$).

2.2 P3 — free-statistic sufficiency (NF-κB spatial gradient)

P3-1: intermediate (1/2 units pass the $\tau(3)$-AUC ≥ 0.60 line); P3-2: **hit** (2/2 units ≥ 0.55); P3-3: **falsified** (0/2 units, both below line). Frozen lines and per-unit values in `P3_verdict_summary.csv`.

2.3 P4 — ERK re-adjudication at trajectory level (GPCR-KTR)

Clause	Frozen line	Value	Verdict
P4-1	4 units $\tau(3)$ AUC ≥ 0.60; all below $\rightarrow$ falsified	all 8 AUCs < 0.60 (max 0.5819; one 0.4695)	**Falsified**
P4-2	hit iff $\text{AUC}_H < 0.60$ and $D \geq 0.05$	$D = 0.0468$, 95% CI (0.0316, 0.0622); $\text{AUC}_L = 0.567$, $\text{AUC}_H = 0.520$	Intermediate
P4-3	hit iff $\geq 8/10$ units $\lvert\delta\rvert \leq 0.01$; falsified iff $\geq 3/10 \geq 0.03$	5/10 units $\lvert\delta\rvert \geq 0.03$	**Falsified**

All 25 per-unit rows (Histamine and UK ladders, low/mid/high/full bands) with CIs are in `P4_verdict_summary.csv`.

2.4 P6 — FCD attribution (NF-κB sequential grid)

$\Delta R^2 = R^2_{\text{fold}} - R^2_{\text{abs1}} = -0.0014$, 95% CI $(-0.252, 0.153)$, 18 units: **intermediate** ($R^2_{\text{fold}} = 0.0063$, $R^2_{\text{abs1}} = 0.0078$, fold slope -0.0086). The self-built arm ($\Delta R^2 = -0.178$) is a non-scoring control under the pre-registered divergence rule. Verbatim in `P6_verdict_summary.csv`.

2.5 P7 — FCD–Weber adjudication (archived copy)

First pass (25 units): $\Delta R^2 = +0.118$, CI $(-0.296, 0.468)$, intermediate; erratum re-adjudication (30 units): $\Delta R^2 = -0.091$, CI $(-0.445, 0.277)$, intermediate. The chemotaxis companion repository holds the full P7 pipeline and unit tables.

3 Morphogen instance: the Bcd absolute anchor (P9)

Per-embryo Bicoid profiles (Nikolić et al. repository; Liu et al. PNAS 2013 data), $n = 581$ embryos:

Quantity	Value	95% CI
Profile-length scaling exponent α (P9-1)	-0.182 (s.e. 0.143)	archived in JSON
Theil–Sen slope	-0.199	—
Median λ (μm)	87.6	—
Boundary exponent α_{boundary} (P9-2, $n = 609$)	-0.644 (s.e. 0.505)	archived in JSON
γ_{B0} (P9-3)	0.379	archived in JSON
Upgrade arm α_b ($n = 301$)	1.942 (s.e. 0.755)	archived in JSON

Embryo-length range archived in `p9_bcd_anchor_results.json`. Axes-1/2 of the Bergmann 2010 Dataset-S1 audit are computed by `audit_bergmann2010_axis12.py` (output archived in `records/logs/`).

4 Cascade instance: per-experiment Hill fits (GPCR→Ca)

Fits of the per-experiment dose–response (top, EC_{50}, Hill n) on the Keshelava 2018 per-cell peak tables, two fitting ladders (default and Methods) archived in `cascade_audit_results.json` and `cascade_hill_fits_methods_ladder.json`; the two agree to $\lesssim 10^{-3}$ throughout (Table 2).

The source archive contains 27 experiments (numbered 01–29 with 08 and 12 absent); 26 yielded a usable fit, experiment 27 did not. Over these 26 fits the median EC_{50} is 0.632 μM, the coefficient of variation 1.161, and the range 0.159–5.165 μM (32.4-fold); these are the values quoted in the main text.

5 P10: gain-distribution confinement across systems

Confinement ratio R = within-cell CV (median) / across-cell CV, per condition (KTR archive of Chavez-Abiega et al. 2022; NF-κB archive of Wang et al. 2022):

Split-half rank correlations and 95% CIs per condition are archived in `p10_gain_confinement_results.json`. Supplementary arms (`p10_supplementary_arms_results.json`): S1P/CN ERK median $R = 2.17$ (2 conditions); Histamine/UK Akt-RB arms have no qualifying conditions ($n = 0$).

6 Tumour chain (P11–P16)

6.1 P11: counting law across cell lines

Digitised arms archived in `tumour_chain_P11_P16/P11_digitized_*.csv`: Finzel dissertation Fig. 30 (per-line FWHM medians and quartiles at 2.5/5/10 Gy; e.g. A549 107 min at 2.5 and 5 Gy, 122 min at 10 Gy) and Stewart-Ornstein et al. 2017 (per-line periodicity at 1 vs 8 Gy, ATM IC_{50}, γH2AX fraction remaining at 8 h; 12 lines).

Table 2: Per-experiment Hill fits (Methods ladder), verbatim.

Experiment	top	EC_{50} (μM)	Hill n	n_{cells}
01	0.199	0.368	1.47	9
02	0.193	0.317	1.91	26
03	0.254	0.447	1.51	14
04	0.276	0.376	2.65	12
05	0.195	0.402	1.41	10
06	0.182	0.637	1.34	17
07	0.193	0.317	1.91	26
09	0.330	0.323	2.98	12
10	0.258	0.693	2.14	12
11	0.204	0.869	1.41	11
13	0.182	0.627	1.29	16
14	0.227	1.227	1.83	3
15	0.278	0.388	2.61	11
16	0.293	1.678	3.82	10
17	0.266	0.758	2.73	9
18	0.169	0.775	1.77	14
19	0.191	0.565	2.85	8
20	0.163	0.750	1.34	9
21	0.206	0.749	1.85	29
22	0.269	5.165	0.93	10
23	0.145	0.783	1.23	18
24	0.107	0.159	2.25	1
25	0.154	0.375	1.47	10
26	0.191	0.532	1.82	18
28	0.229	0.795	1.90	16
29	0.297	0.685	1.97	22

6.2 P12b: resistance mutations and the zero direction

EGFR saturation-mutagenesis audit (GSE305057, 1380 missense variants tested; full per-variant table `p12b_egfr_dms_variants.csv`, 93 KB):

Drug-specific arms (`p12b_egfr_dms_results.json`): osimertinib 479 vs 901 variants; BLU-945 128 vs 1252 (median ratio 0.122). ABL catalytic axis (EVER study: Liu et al., *Commun. Biol.* **3**, 30 (2020), Supplementary Data 1, PMC6952392; our table `P12b_abl_catalytic_efficacy.csv`): WT $k_{cat}/K_M = 495.4$; L248R fold 0.05 ($|\Delta \log_{10}| = 1.30$, ambiguous-sensitivity class); Y253H fold 2.46 (clinical class); extended clinical expansion in `P12b_extended_clinical_expansion.csv` and `P12b_final_expansion.csv` (incl. E255V, T315I gatekeeper excluded).

6.3 P13: channel capacity across Ras/RAF genotypes

Mutual information I(input; ERK output), Miller–Madow corrected, per ligand × genotype (bits; Δ vs KrasWT), computed on the single-cell EKAR trajectories of Gillies et al., *Mol. Syst. Biol.* **16**, e9518 (2020), Source Data Figs 2 and 6 (PMC7569415):

(*G12V AREG $I = 0.093$.) Per-genotype n (1116–2522 cells) and 95% bootstrap CIs in

System	Condition	n cells	within-CV med.	R
ERK	Histamine 2.33 μM	41	0.565	0.775
ERK	Histamine 100 μM	35	0.700	0.410
ERK	UK 33.3 μM	39	0.278	1.000
ERK	UK 100 μM	59	0.250	1.160
NF-κB	pooled	11,265	2.520	1.203

Arm	n	median activity	ratio vs all missense	one-sided MW p
Curated resistance	17	0.01609	2.22	0.0022
Clinical resistance	6	0.01614	2.23	0.096

`p13_mi_results.json` (six ligands). Robustness (`p13_robustness.json`): the ordering is stable across 4/6/8/12/16 histogram bins and under the KSG estimator (e.g. BRaf-V600E Δ: -0.397 to -0.423 across bins; KSG -0.421). Response-decay times (`p13_fig6_decay.json`): median t_0 per genotype 2.76–6.23 h (KrasWT 6.23, HRasWT 2.95, NrasWT 5.14, G12C 5.19, G12D 4.65, G12V 2.76, Q61R 2.85, BRaf-V600E 5.32).

6.4 P14: downstream shape decoding

Data: Jimenez-Asins et al., *Mol. Syst. Biol.* **18**, e10588 (2022), RNA-seq TPM and TMT proteomics from the authors' public repository (`albajimenezasins/Proteomics_MSB_2022`). 9,103 induced genes; per-gene pulse-vs-sustained log-ratio table `p14_pergene_log2R.csv` (671 KB). Fraction of genes with sustained $>2\times$ pulsed: 29.0%; pulsed $>2\times$ sustained: 11.1%; within $2\times$: 49.3%. Median onset-time CV 0.88. p53-protein AUC per trace: Pulse A 20.78, Pulse B 19.12, Sustain C 36.7, Sustain D 29.18. The 23 fate-gene list is archived in `p14_results.json`.

6.5 P15: TCGA tissue-level decoupling

pERK-vs-ligand-receptor slopes per mutation group, four PanCanAtlas 2018 studies (verbatim from `p15_tcga_results.json`):

Slope ratios (mutant vs WT): COADREAD 4.67× (strong), 3.75× (G12); LUAD 1.75×; PAAD 1.13×; SKCM $-1.81\times$. Interaction terms: COADREAD $\beta = -0.454$ ($p = 0.205$), SKCM $\beta = -0.086$ ($p = 0.802$). Per-sample tables (sample-level g-scores, pEGFR, EGFR, pERK, group) archived as `p15_<study>_persample.csv`.

7 Wnt digitisation and yeast positive control

`fig4F_digitized.csv`: our digitisation of Goentoro & Kirschner (2009) Fig. 4F (7.2 mM: 0.964; 20.8 mM: 1.040; 41.3 mM: 1.021 fold-change; registered precision ± 0.05 fold-change, ± 2 mM). The Hog1 osmotic set-point positive control (`score_markers_v2_authordata.py`) runs against the author numerical table `data_external/muzzey2009/mmc2.xls`; only the glycerol arms rely on figure digitisation (Methods).

Ligand	KrasWT	G12C	G12D	G12V	Q61R	BRaf-V600E
EGF	0.434	−0.201	−0.228	−0.050	−0.335	−0.413
AREG	0.089	−0.078	−0.073	+0.004*	archived in JSON	

Study	Group	n	slope	s.e.	p
COADREAD (n=464)	WT	218	−0.124	0.152	0.416
	KRAS-G12	115	−0.464	0.246	0.062
	strong	57	−0.578	0.333	0.089
	other-mut	74	+0.062	0.361	0.863
LUAD (n=360)	WT	232	−0.255	0.092	0.006
	KRAS-G12	92	−0.445	0.211	0.038
SKCM (n=329)	WT	70	+0.031	0.313	0.923
	strong	198	−0.055	0.174	0.752
PAAD (n=122)	WT	43	−0.274	0.562	0.629
	KRAS-G12	71	−0.310	0.353	0.382

8 Third-party data provenance

Shipped here (small, license-clear): Bergmann 2010 Datasets S1/S2 (PMC2858443, CC BY); Muzzey et al. 2009 `mmc2.xls` (PMC3109981). Referenced but not re-hosted (exact access notes in `data_external/PROVENANCE.md`): Nikolić et al. repository (arXiv:2312.17684) / Liu et al. PNAS 2013; Gregor et al. 2005, 2007; Petkova et al. 2019; Cheung et al. 2014; Chavez-Abiega et al. 2022 KTR archive (∼786 MB); Wang et al. 2022 NF-κB archive (∼54 MB); Keshelava et al. 2018 Suppl. Data 3 (mirrored in the companion repository `gpcr-field-audit`); TCGA PanCanAtlas 2018 via cBioPortal; GSE305057; Gillies et al. 2020, *Mol. Syst. Biol.* **16**, e9518, Source Data Figs 2/6; Liu et al. (EVER), *Commun. Biol.* **3**, 30 (2020), Suppl. Data 1; Finzel Pérez dissertation (FU Berlin 2016, refubium fub188/7115, doi:10.17169/refubium-11314); Jimenez-Asins et al. 2022, *Mol. Syst. Biol.* **18**, e10588 (authors' GitHub repository).